🌐 Module 05: Web Application Penetration Testing

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# Exercise 21: Test for Weak Cryptography Vulnerabilities using AI
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Scenario

Testing web applications for weak cryptography and transport layer security misconfigurations is an essential phase of web application penetration testing. Cryptographic vulnerabilities—such as support for deprecated SSL/TLS protocols (SSLv2, SSLv3, TLS 1.0, TLS 1.1), weak cipher suites (e.g., RC4, 3DES, EXPORT ciphers), and unencrypted data transmission—expose sensitive user traffic to eavesdropping and Man-in-the-Middle (MitM) attacks.

Generative AI platforms (such as ChatGPT) can be used to generate automated bash scripts that streamline SSL/TLS testing tools (e.g., `sslscan`, `testssl.sh`, or `nmap` SSL scripts), combining transport security evaluations into a single execution workflow.

In this exercise, we generate and run `TestWeakCryptography.sh` against target domain **www.certifiedhacker.com**, logging transport layer auditing results (`Transport_Layer_Security_Results.txt`) to determine the target host's IP address (**162.241.216.11**).

Objective

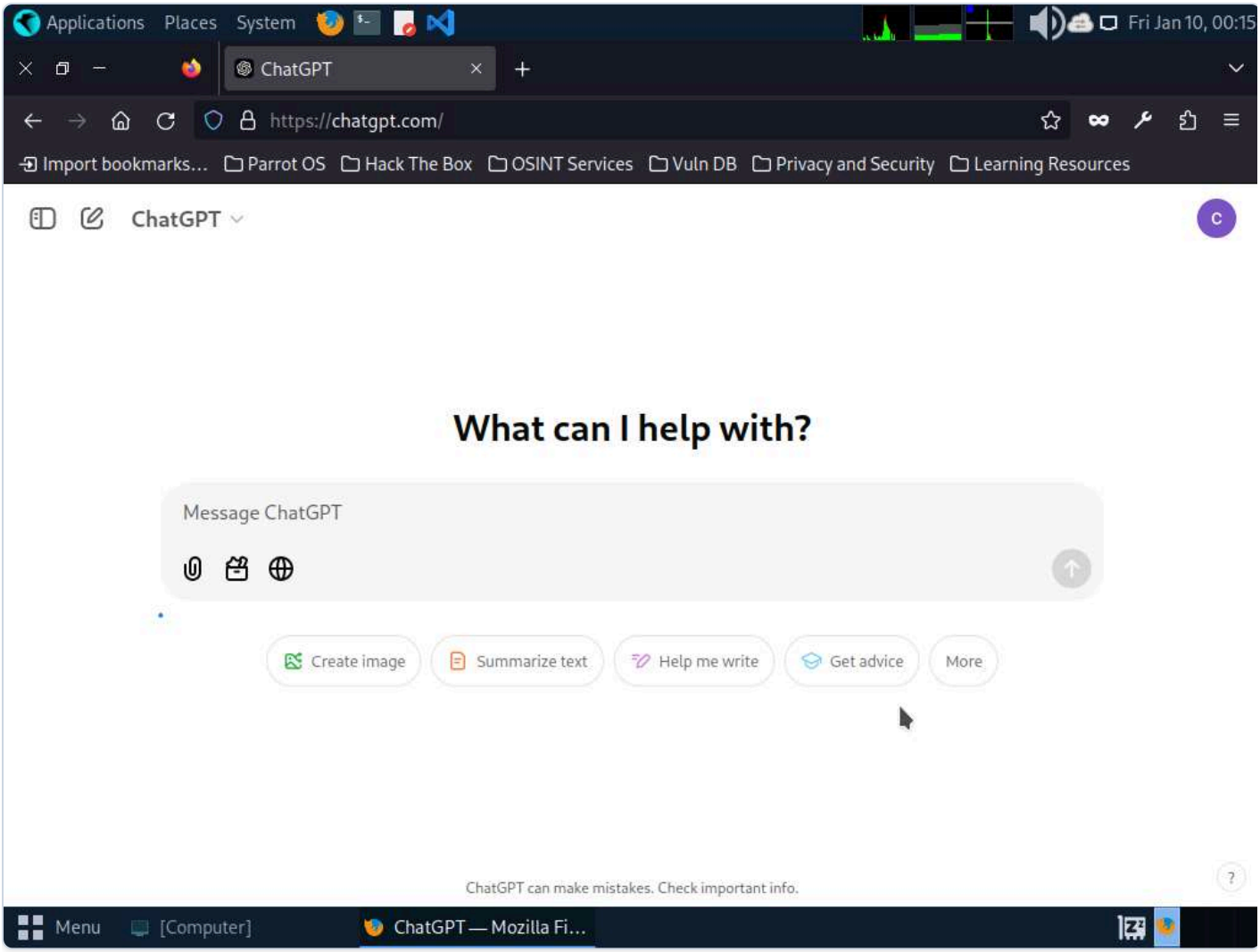
The objective of this lab is to help students learn how to:
- Leverage AI tools (ChatGPT) to generate shell scripts automating cryptographic vulnerability assessments
- Audit target web applications for insufficient transport security, weak ciphers, and unencrypted transmission
- Execute Unix-formatted shell scripts (`TestWeakCryptography.sh`) using `dos2unix` and `chmod`
- Analyze output logs (`Transport_Layer_Security_Results.txt`) to extract target host IP resolution (**162.241.216.11**)

Task 1: Test for Weak Cryptography Vulnerabilities using AI

In this task, we use ChatGPT on Parrot Security to generate a cryptographic auditing script, execute `TestWeakCryptography.sh` located in `/home/pentester/Scripts/Module 5` against target domain **www.certifiedhacker.com**, and review the resulting `Transport_Layer_Security_Results.txt` report to identify the target IP address (**162.241.216.11**).

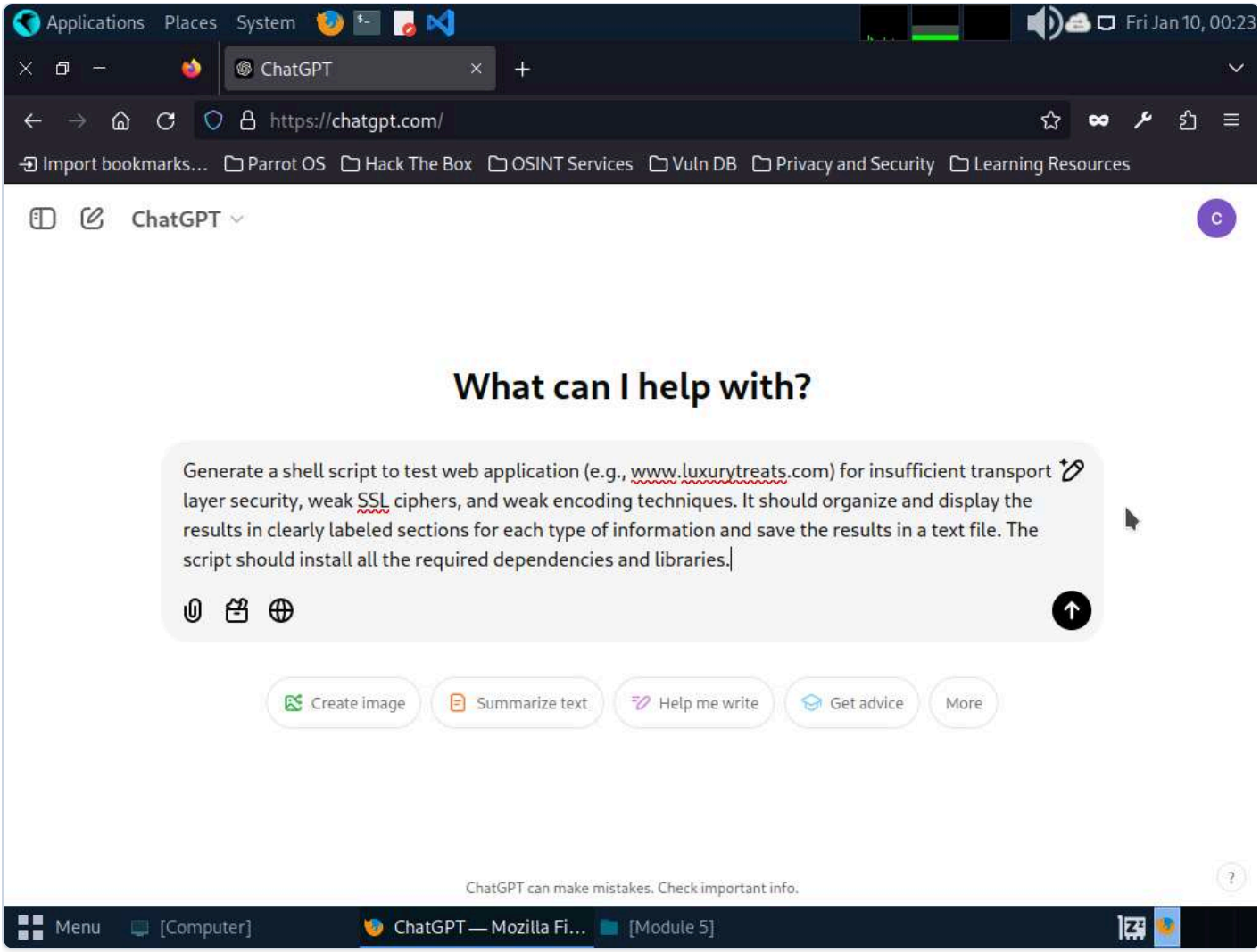
Lab Tasks

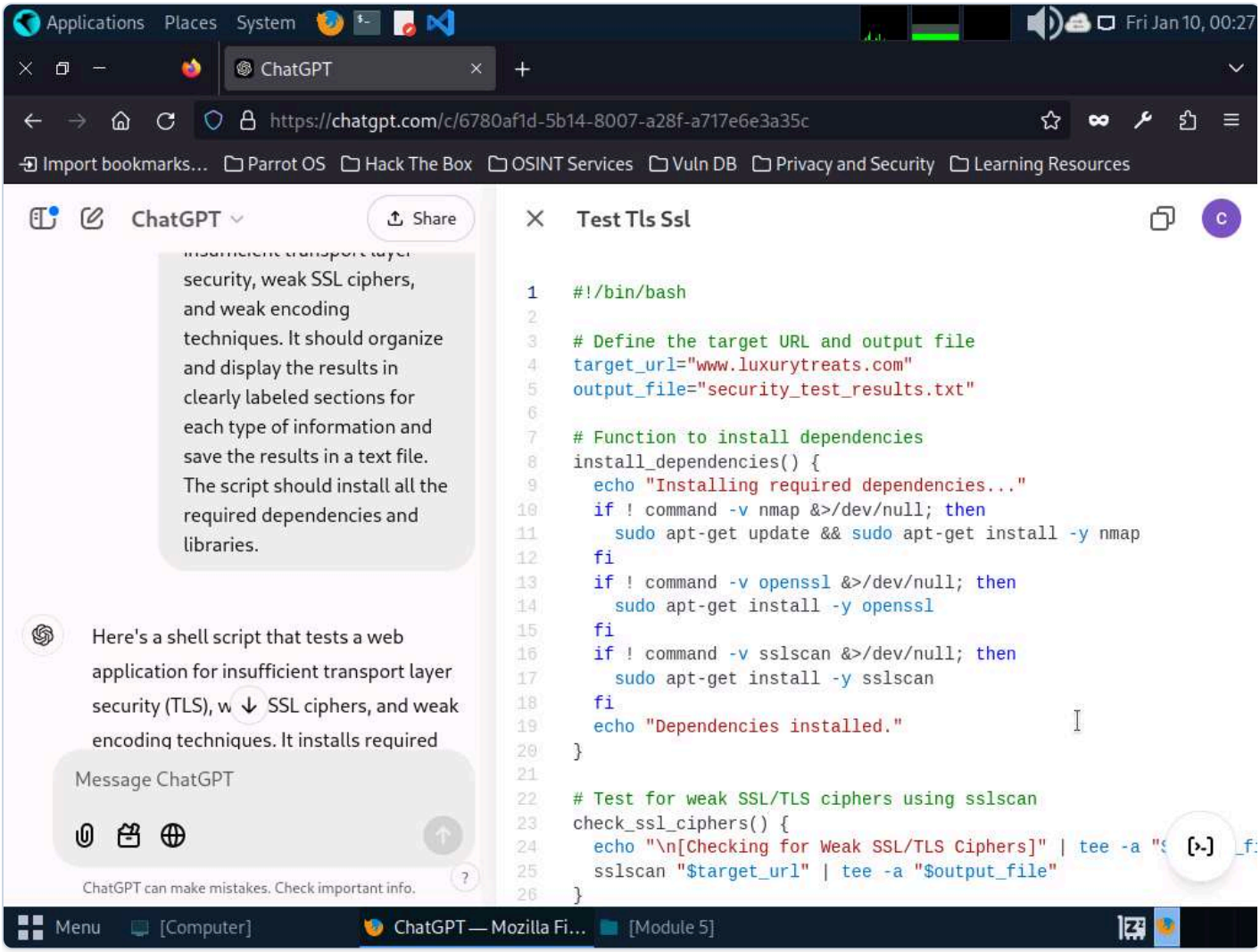
1. In **Parrot Security**, open Firefox and navigate to **https://chatgpt.com/**.



1. Log into ChatGPT using your Google account if not already authenticated.
2. Submit the following prompt in the ChatGPT message box to generate the script:

> *"Generate a shell script to test web applications (e.g., www.luxurytreats.com) for insufficient transport layer security, weak SSL ciphers, and weak encoding techniques. It should organize and display the results in clearly labeled sections for each type of information and save the results in a text file. The script should install all the required dependencies and libraries."*


Applications Places System
Fri Jan 10, 00:23
ChatGPT
https://chatgpt.com/
Import bookmarks...
Parrot OS
Hack The Box
OSINT Services
Vuln DB
Privacy and Security
Learning Resources
ChatGPT
C
What can I help with?
Generate a shell script to test web application (e.g., www.luxurytreats.com) for insufficient transport layer security, weak SSL ciphers, and weak encoding techniques. It should organize and display the results in clearly labeled sections for each type of information and save the results in a text file. The script should install all the required dependencies and libraries.
Create image
Summarize text
Help me write
Get advice
More
ChatGPT can make mistakes. Check important info.
Menu
[Computer]
ChatGPT — Mozilla Fi...
[Module 5]



1. Review the generated script structure:

```
#!/bin/bash
# TestWeakCryptography.sh - Automated Cryptographic Vulnerability Audit

echo "================================================"
echo " Cryptographic & Transport Layer Security Audit   "
echo "================================================"

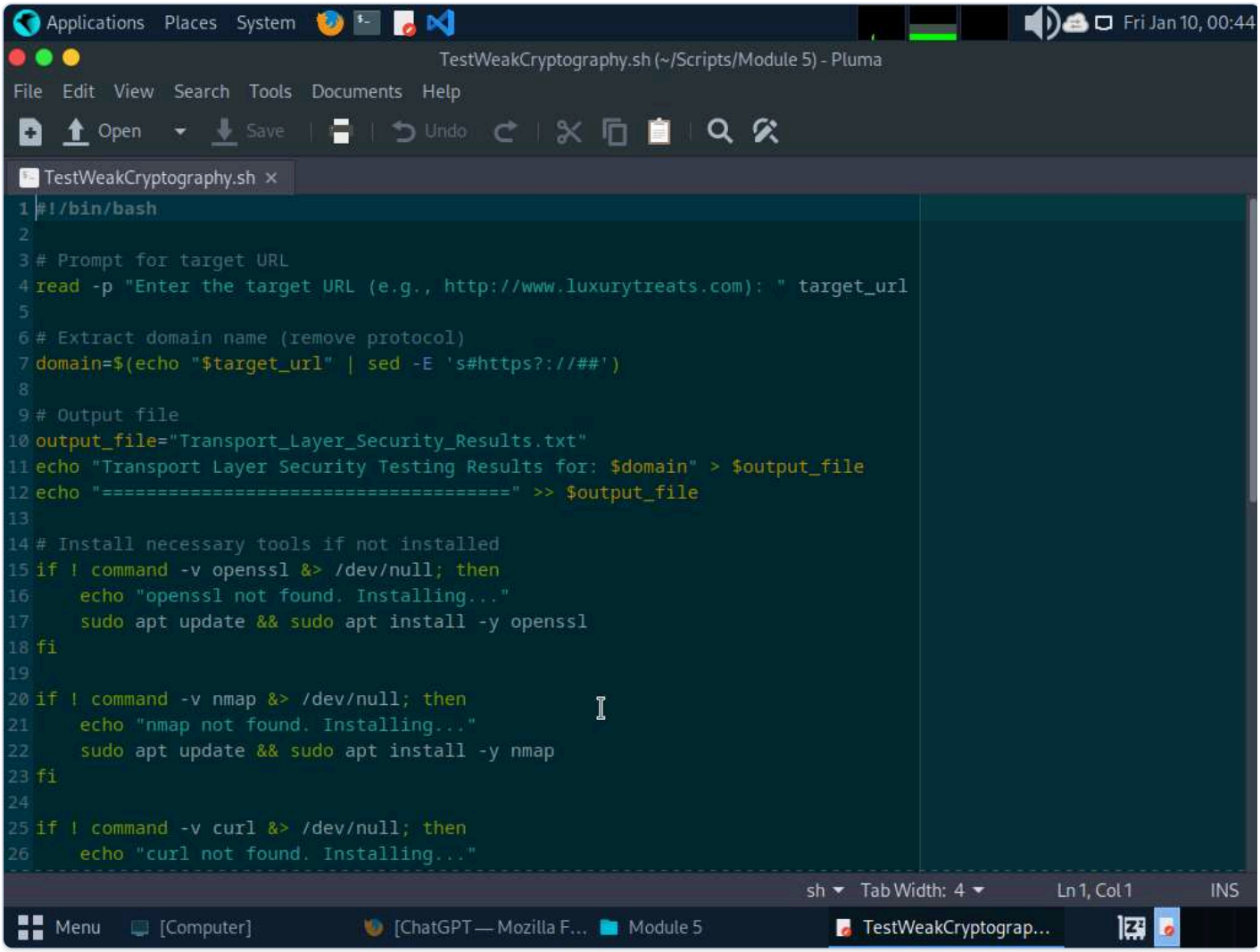
read -p "Enter Target Domain/URL (e.g., www.certifiedhacker.com): " TARGET_HOST
OUTPUT_FILE="Transport_Layer_Security_Results.txt"

# Install dependencies if missing
apt-get update && apt-get install -y sslscan nmap

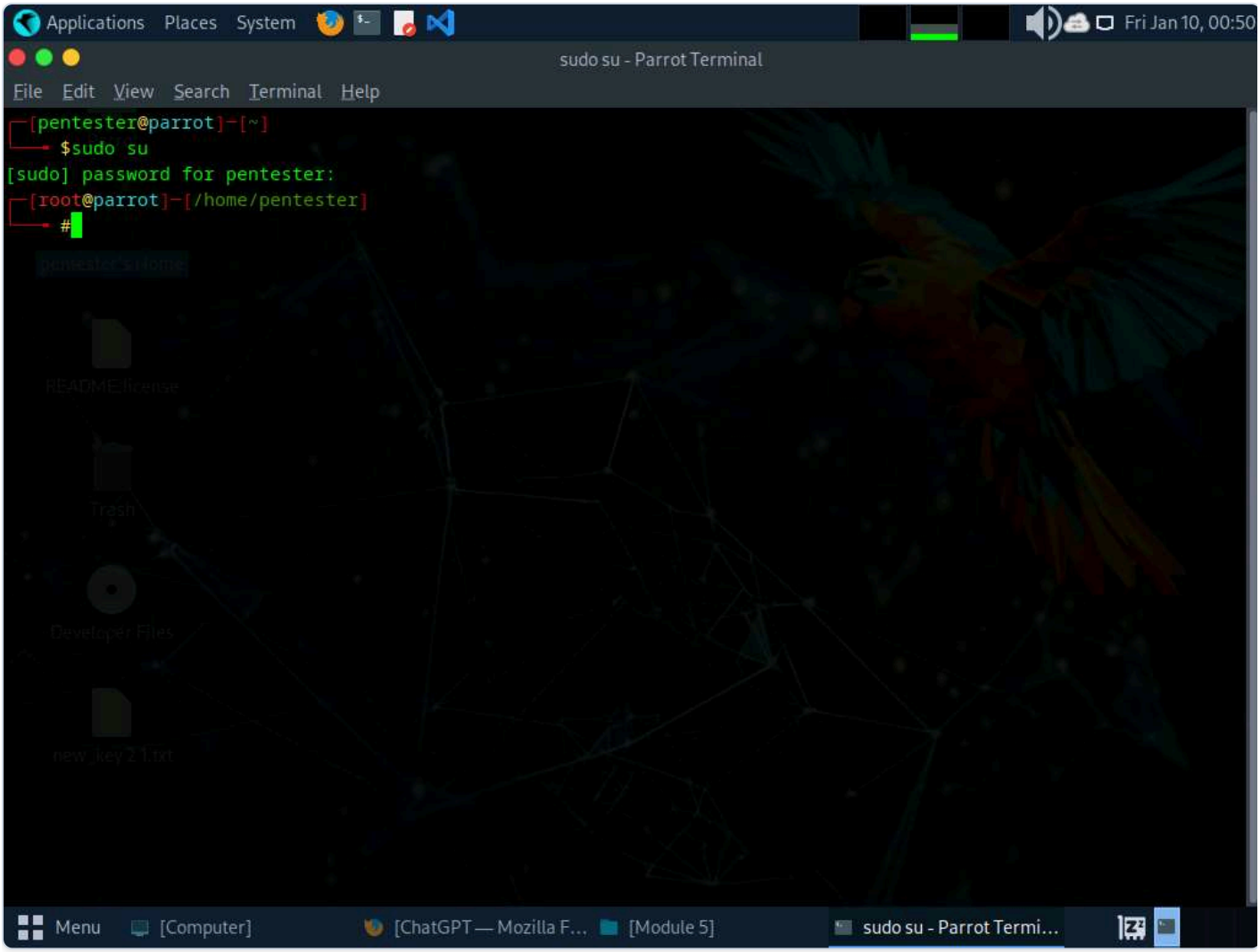
echo "Auditing Transport Layer Security & SSL/TLS Ciphers..." | tee -a $OUTPUT_FILE
sslscan $TARGET_HOST | tee -a $OUTPUT_FILE

echo -e "
Running Nmap Cipher Enum Scripts..." | tee -a $OUTPUT_FILE
nmap --script ssl-enum-ciphers -p 443 $TARGET_HOST | tee -a $OUTPUT_FILE

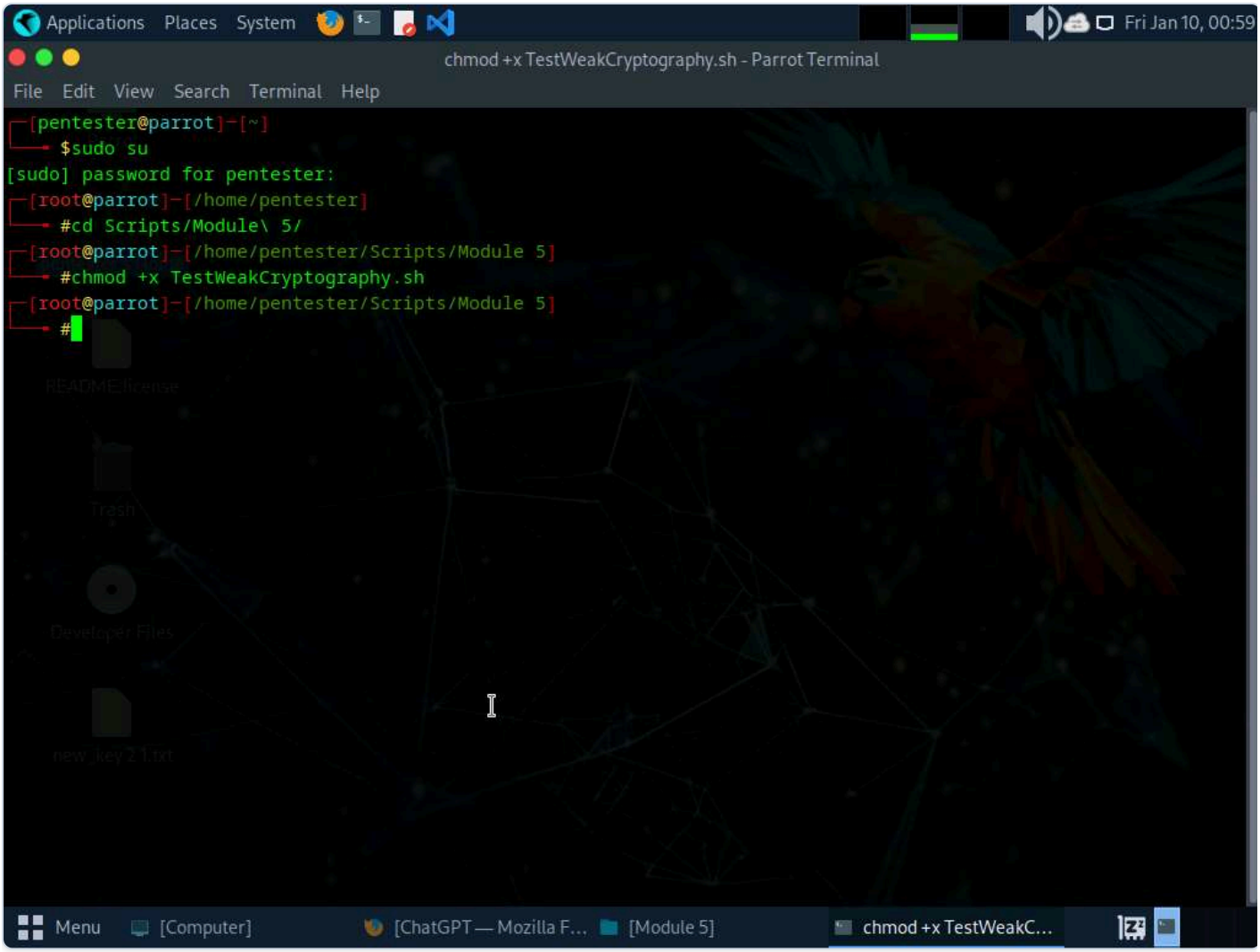
echo -e "
Audit Complete. Output saved to $OUTPUT_FILE."
```



1. Open a terminal with root privileges (**sudo su**, password: **toor**).

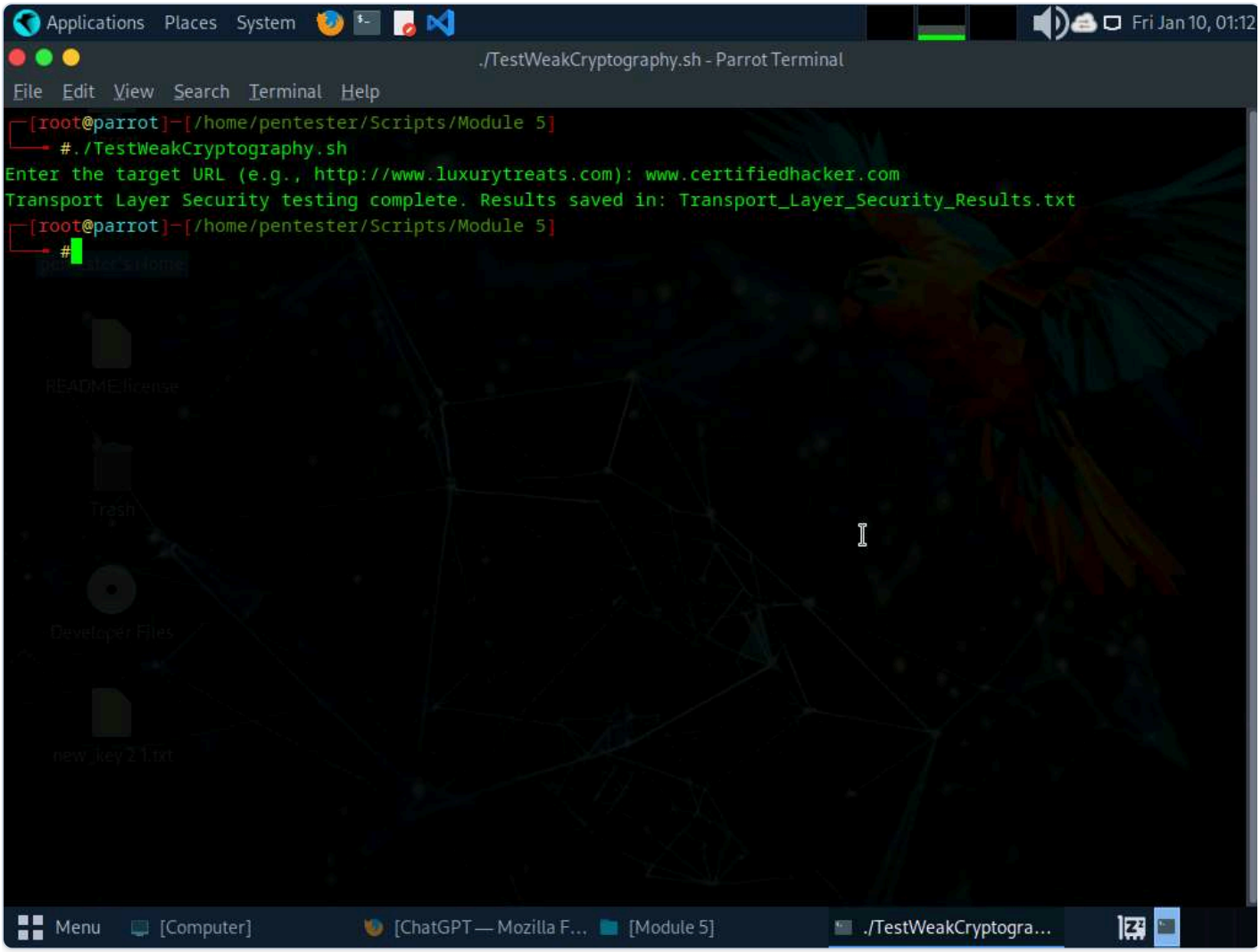


1. Navigate to the pre-saved script directory (**/home/pentester/Scripts/Module 5**), convert line endings, and grant execution permissions:
   cd "/home/pentester/Scripts/Module 5" && dos2unix TestWeakCryptography.sh && chmod +x TestWeakCryptography.sh

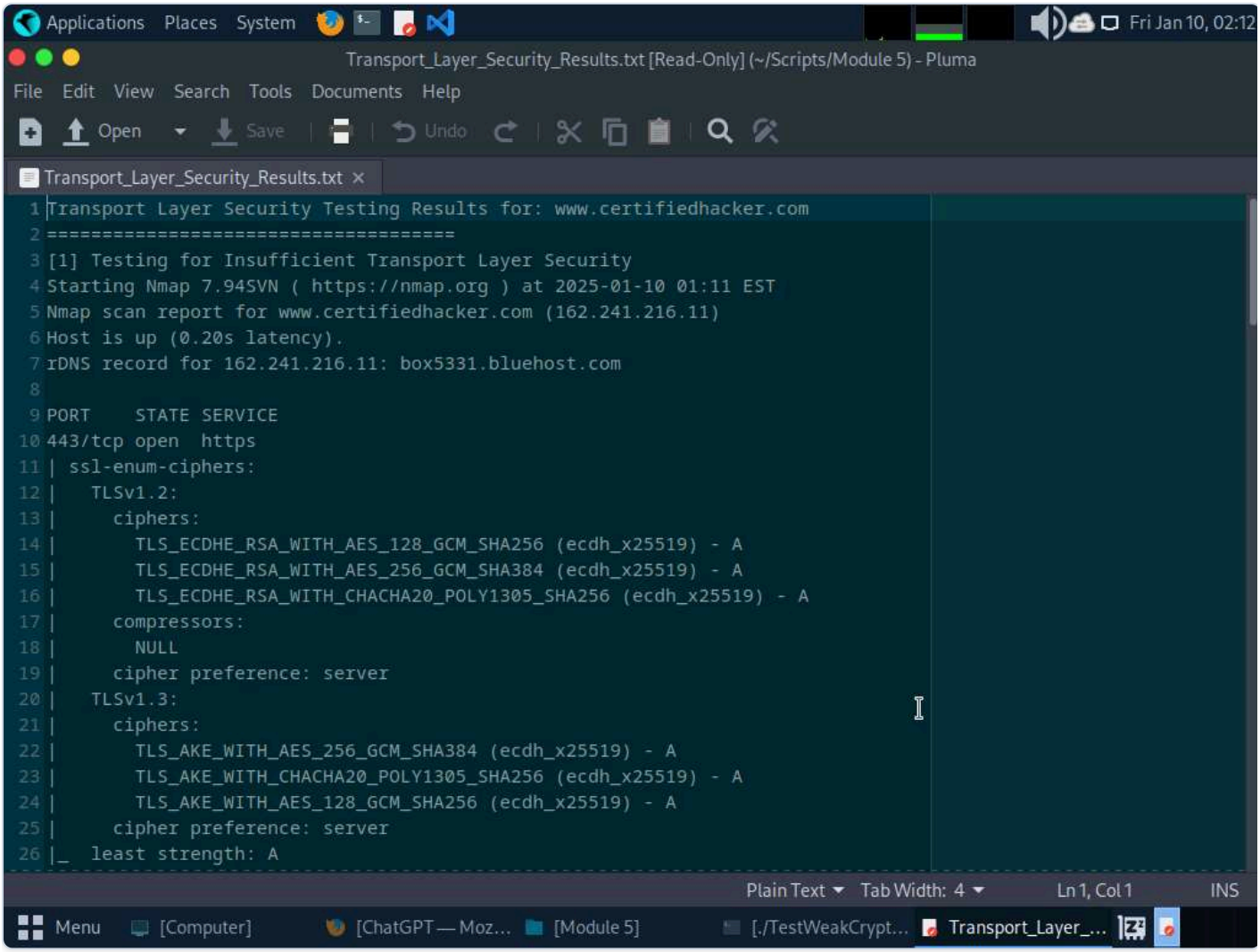


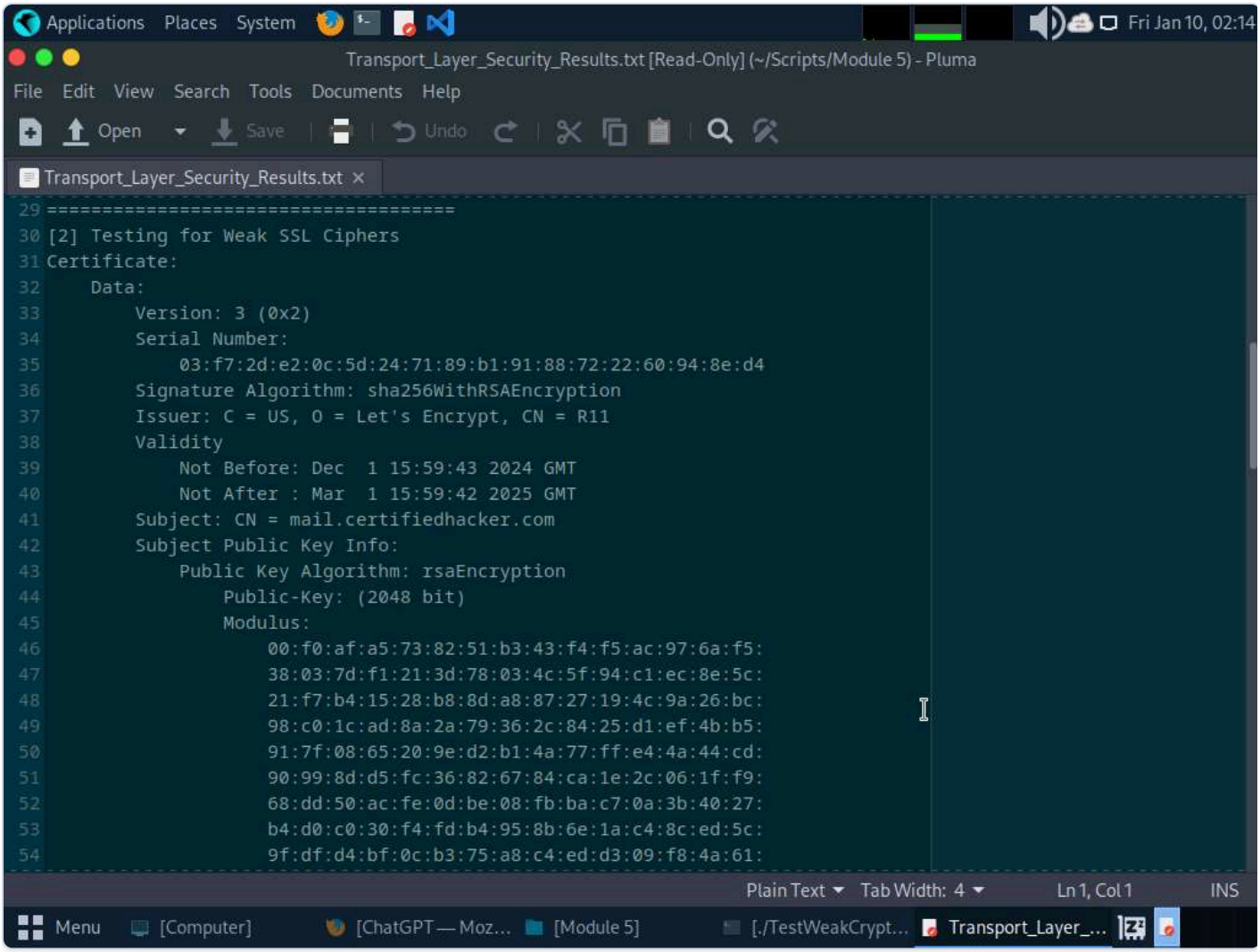
1. Run the script:
   ./TestWeakCryptography.sh

2. When prompted for the target URL, type **www.certifiedhacker.com** and press Enter.



1. Upon script completion, open and review the results file (**/home/pentester/Scripts/Module 5/Transport_Layer_Security_Results.txt**).


Applications Places System
Fri Jan 10, 02:12
Transport_Layer_Security_Results.txt [Read-Only] (~/Scripts/Module 5) - Pluma
File Edit View Search Tools Documents Help
Open
Save
Undo
Transport_Layer_Security_Results.txt
Transport Layer Security Testing Results for: www.certifiedhacker.com
==========
[1] Testing for Insufficient Transport Layer Security
Starting Nmap 7.94SVN (https://nmap.org) at 2025-01-10 01:11 EST
Nmap scan report for www.certifiedhacker.com (162.241.216.11)
Host is up (0.20s latency).
rDNS record for 162.241.216.11: box5331.bluehost.com
PORT STATE SERVICE
443/tcp open https
| ssl-enum-ciphers:
| TLSv1.2:
| ciphers:
| TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256 (ecdh_x25519) - A
| TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384 (ecdh_x25519) - A
| TLS_ECDHE_RSA_WITH_CHACHA20_POLY1305_SHA256 (ecdh_x25519) - A
| compressors:
| NULL
| cipher preference: server
| TLSv1.3:
| ciphers:
| TLS_AKE_WITH_AES_256_GCM_SHA384 (ecdh_x25519) - A
| TLS_AKE_WITH_CHACHA20_POLY1305_SHA256 (ecdh_x25519) - A
| TLS_AKE_WITH_AES_128_GCM_SHA256 (ecdh_x25519) - A
| cipher preference: server
|_ least strength: A
Plain Text
Tab Width: 4
Ln 1, Col 1
INS
Menu
[Computer]
[ChatGPT — Moz...
[Module 5]
[./TestWeakCrypt...
Transport_Layer_...



1. Locate the resolved target IP address disclosures in the scan headers: **162.241.216.11**.
2. Close all open terminal and browser windows.

This concludes the lab exercise.

> **IMPORTANT:** ❓ **Question 5.21.1.1** On a Parrot Security machine, utilize AI tools such as ChatGPT to generate a shell script that automates the process of testing for cryptographic vulnerabilities. This includes identifying weak SSL ciphers, insufficient transport layer security, and insecure encoding techniques on `www.certifiedhacker.com`. Use the IP address of `www.certifiedhacker.com` obtained from the scan as an input for the script. [!SUCCESS] ✅ **Answer:** **162.241.216.11**